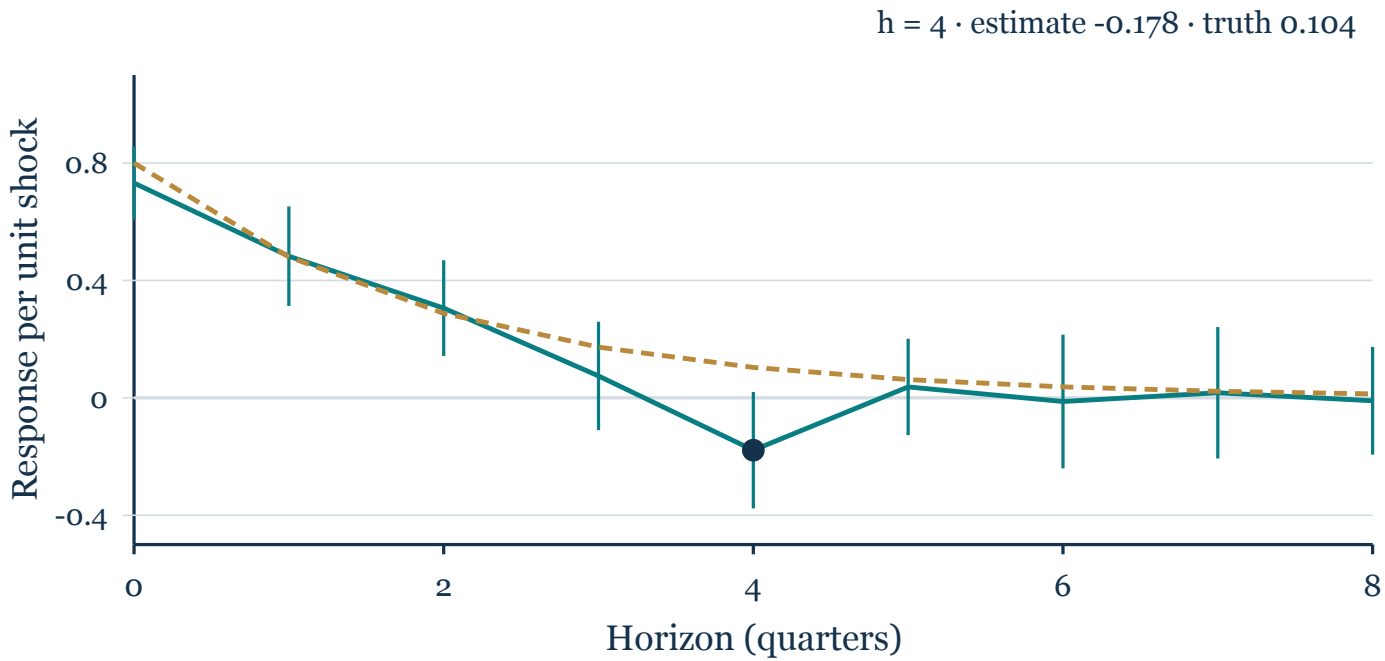
Which change addresses the problem we found?

Compare improving dynamics, changing an error model and estimating a future response.



Constructed quarterly process; actual local-projection estimates with approximate pointwise 95% intervals. The dashed path is the known response.

Recall what HAC preserves

An uncertainty adjustment does not add predictors.

> Newey–West retains the OLS coefficient on the same equation and sample while changing its estimated covariance.

> Before comparing the results, say which concern motivates each operation.

Add a lagged outcome with its timing made explicit

A different coefficient can be a different conditional comparison.

```
. newey logacc L.logacc trend i.month unem spdlaw beltlaw wkends, lag(4)

------------------------------------------------------------------------------
             |             Newey-West
      logacc | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
      logacc |
         L1. |   .3326578   .1071252     3.11   0.003     .1198023    .5455133
             |
       trend |   .0008893   .0002853     3.12   0.002     .0003223    .0014562
             |
        unem |  -.0125834   .0044189    -2.85   0.005    -.0213638   -.0038031
      spdlaw |  -.0414191   .0141509    -2.93   0.004    -.0695366   -.0133016
     beltlaw |   .0652123   .0154671     4.22   0.000     .0344794    .0959452
      wkends |   .0032099    .002814     1.14   0.257    -.0023815    .0088014
       _cons |   7.018421   1.156725     6.07   0.000     4.720033    9.316809
------------------------------------------------------------------------------
```

Calendar-indicator coefficient rows are omitted here. The output download contains the full numerical results.

Dynamic completeness is a conditional-mean claim

Past information should not predict the remaining mean error.

> A dynamically complete mean specification uses the relevant history so additional past information does not improve its conditional-mean prediction.

> For accidents, past travel conditions or enforcement changes may be relevant.

Prais–Winsten models an AR(1) error process

Its additional structure can change the fit.

```
. prais logacc trend i.month unem spdlaw beltlaw wkends

------------------------------------------------------------------------------
      logacc | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
       trend |   .0009183    .000368     2.50   0.014     .0001874    .0016493
             |
        unem |   -.019975   .0047144    -4.24   0.000    -.0293397   -.0106104
      spdlaw |   -.045256   .0179614    -2.52   0.013     -.080934   -.0095779
     beltlaw |   .1008944   .0201055     5.02   0.000     .0609572    .1408315
      wkends |   .0042874   .0028062     1.53   0.130    -.0012867    .0098615
       _cons |   10.61796   .0628171   169.03   0.000     10.49318    10.74274
-------------+----------------------------------------------------------------
```

Calendar-indicator coefficient rows are omitted here. The output download contains the full numerical results.

A horizon

Apply the idea before reading the solution.

Practice

In quarterly local projections $Y_{t+h} = a_h + b_h\, shock_t + \text{controls} + u_{t+h}$, what does b_4 describe?

A horizon: explanation

Connect the calculation to the empirical claim.

> It describes the outcome association or response four quarters, one year, after a unit shock, conditional on the specification. A causal impulse-response interpretation requires valid shock identification and controls; the regression form alone does not provide it.

A local projection asks about a specified future outcome

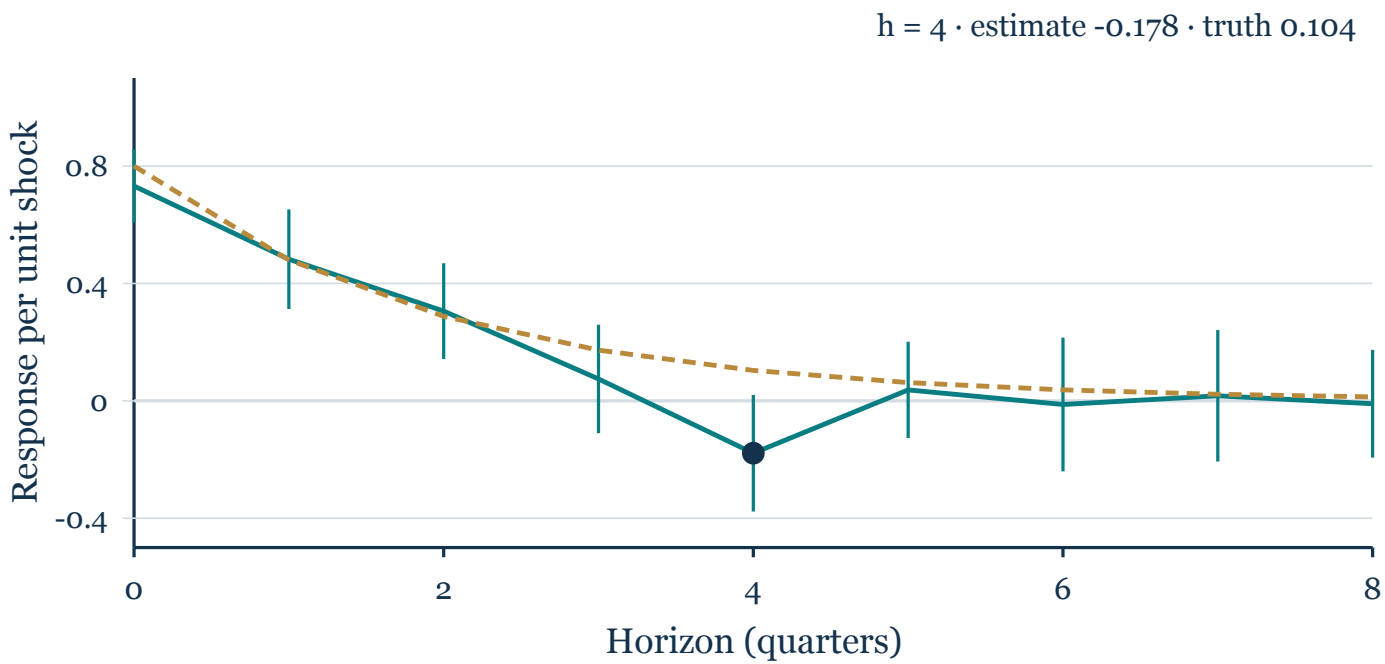
Define the shock and horizon before the coefficient.

> $\pi_{t+h} = a_h + b_h s_t + c_h \pi_{t-1} + u_{t+h}$

> π: inflation; s: shock; h: quarters ahead; b_h: response

Specify the known quarterly data-generating rule

A stable process gives a transparent response benchmark.



Constructed quarterly process; actual local-projection estimates with approximate pointwise 95% intervals. The dashed path is the known response.

- $\pi_t = 0.6\pi_{t-1} + 0.8s_t + \varepsilon_t$
- True impulse response: $b_h = 0.8(0.6)^h$

The horizon coefficient has outcome-per-shock units

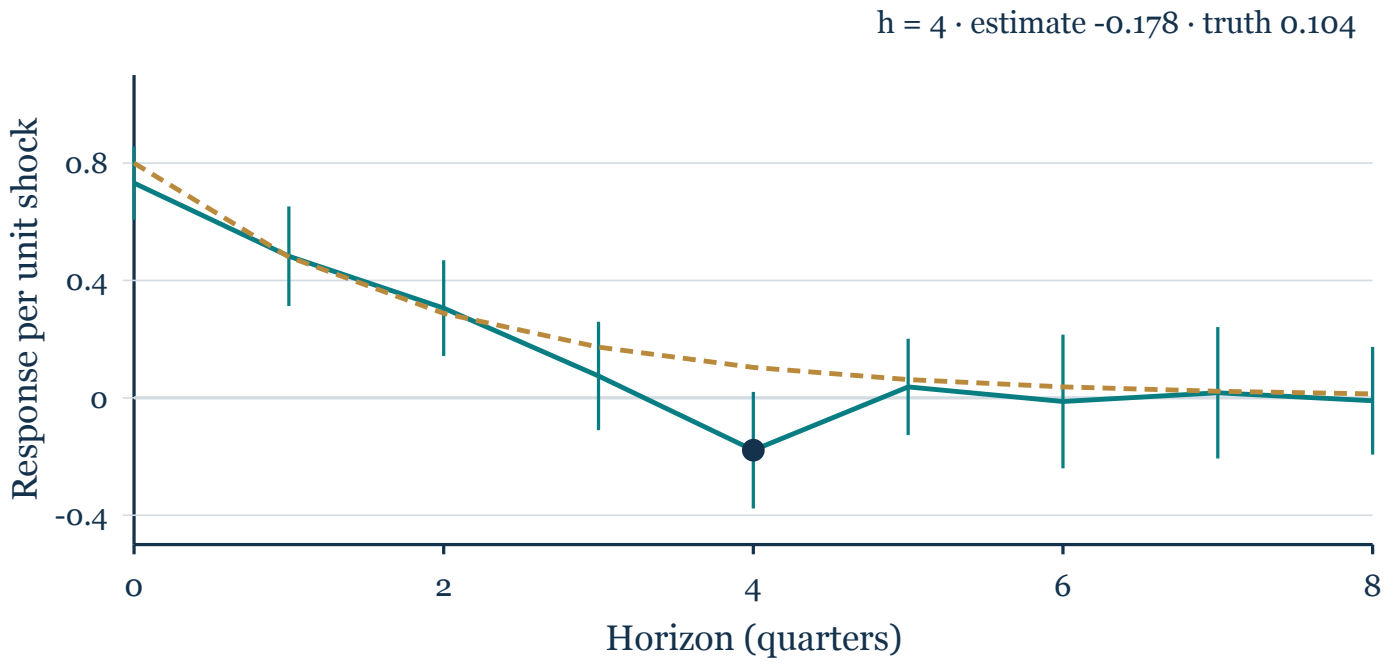
Do not confuse a future value with a cumulative sum.

> At horizon four, the coefficient concerns inflation four quarters after the shock.

> Write the economic sentence before calculating: a one-unit shock is associated with a fitted difference in the outcome at a specified future date, conditional on the included information.

Inspect the estimated path without forcing it to match the truth

Finite-sample response estimates can be noisy.



Constructed quarterly process; actual local-projection estimates with approximate pointwise 95% intervals. The dashed path is the known response.

Read the horizon-four output honestly

A negative sample estimate can coexist with a positive true response.

```
. newey F4.inflation shock L.inflation, lag(8)

------------------------------------------------------------------------------
             |             Newey–West
F4.inflation | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
       shock |    -.17841   .1013114    -1.76   0.080    -.3780179    .0211979
             |
   inflation |
          L1. |   .0652363   .0710399     0.92   0.359    -.0747295    .2052021
             |
       _cons |  -.2784281   .1669179    -1.67   0.097    -.6072968    .0504406
------------------------------------------------------------------------------
```

Future-outcome errors can overlap across dates

A sensible mean model can still leave serial dependence.

> For a horizon longer than one, future outcomes from neighboring starting dates can be affected by some of the same intervening shocks.

> The overlap argument concerns the regression errors at a specified horizon.

Model or inference?

Apply the idea before reading the solution.

Practice

Compare adding a lagged regressor, changing OLS SEs to HAC, and fitting Prais–Winsten. Which can change coefficients?

Model or inference?: explanation

Connect the calculation to the empirical claim.

> Adding regressors changes the conditional-mean model and generally the OLS coefficients and sample. HAC on the identical specification and sample preserves OLS coefficients. Prais–Winsten transforms/weights the model under a modeled error process and can change coefficients as well as SEs; its assumptions must be justified.

A HAC lag has calendar meaning

The same number describes different windows at different frequencies.

> The simulation uses quarterly data and Newey–West lag eight, allowing contributions through eight quarters, or two years.

> The appropriate window depends on the dependence structure and the asymptotic justification, and a finite-sample analysis should inspect plausible choices.

Pointwise intervals do not cover an entire response path at the same rate

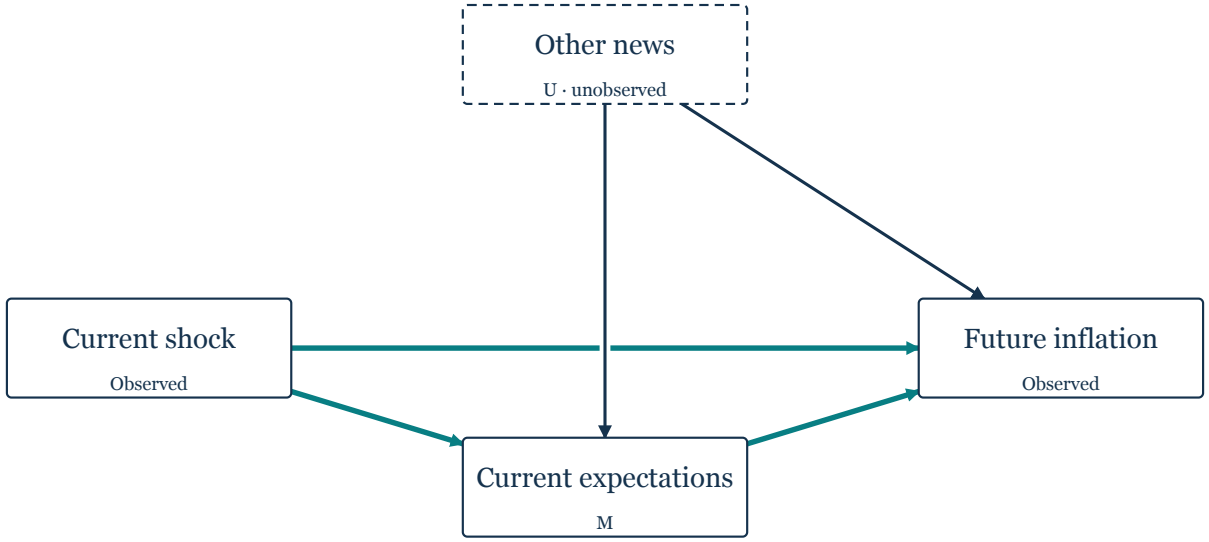
Multiple horizons change the inferential question.

> A pointwise 95% interval refers to one horizon considered on its own under the procedure's assumptions.

> If the research claim concerns a whole response path or a sum, define an inference procedure for that combined quantity.

Controlling a mechanism can change the response being studied

A current mediator is not just another lag control.



2 open directed routes · 0 open alternatives · gold: conditioned on

Assumed causal structure · dashed boxes: unobserved · gold boxes: conditioned on

Initial information and post-shock information have different roles

Timing determines what a control can do.

> A variable known before the shock can help account for the state of the economy that predicts both measured shocks and future outcomes.

> Write a time line for the shock, controls and outcome.

Distinguish an estimated shock from a known innovation

Measurement and first-step uncertainty can matter.

> Our simulation observes its independent shock exactly.

> The appropriate treatment depends on how they were obtained and the inferential target.

Choose among the three changes using the application

The preferred response follows the diagnosed failure.

> For omitted predictable dynamics, reconsider the conditional-mean equation and its timing assumptions.

> None automatically solves policy endogeneity or shock mismeasurement.

Mediation is not proven

Apply the idea before reading the solution.

Practice

A shock coefficient becomes smaller after adding current expectations. Does this prove that expectations mediate the causal effect?

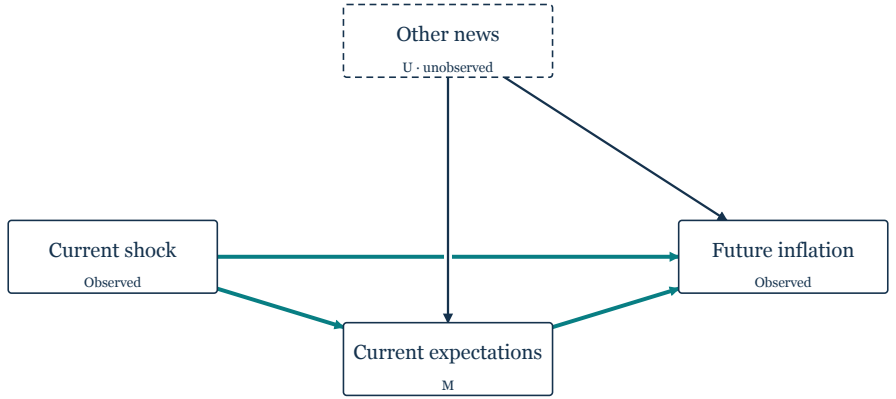
Mediation is not proven: explanation

Connect the calculation to the empirical claim.

> No. It is consistent with a mediation story but may reflect confounding, post-treatment conditioning, endogeneity or other specification changes. State the target and the assumptions under which conditioning has a direct-effect interpretation.

DAG practice: expectations after a shock

Trace the routes before choosing a control.



2 open directed routes · 0 open alternatives · gold: conditioned on

Assumed causal structure · dashed boxes: unobserved · gold boxes: conditioned on

Practice

A shock changes expectations and future inflation. Other news affects expectations and future inflation. Explain why adding current expectations can remove a mechanism and also create an alternative route, even if the shock initially had no common cause with the news.

DAG practice: expectations after a shock: explanation

Explain what conditioning changes in this stated graph.

> Expectations lie on $s_t \to M \to \pi_{t+h}$, so conditioning on them blocks part of the shock's response. They also form a collider on $s_t \to M \leftarrow U \to \pi_{t+h}$; conditioning can open the route through other news. The adjusted shock coefficient therefore does not automatically identify a direct response. An observed reduction in its size is compatible with several stories. State the target, the control's timing and the additional assumptions needed to separate mediation from selection or mediator–outcome confounding.

Compare a known response with a noisy estimate

Reproduce the evidence, then explain its economic meaning.

Practice

Run meeting_12.do. In its explicitly constructed quarterly experiment, calculate the true response four quarters after a unit shock. Compare it with the actual horizon-four estimate and approximate interval. Explain what the discrepancy teaches about finite-sample estimates and why the experiment does not identify an observed historical supply shock.

Compare a known response with a noisy estimate: explanation

A reproducible result needs both a calculation and an interpretation.

> The known response is 0.8 times 0.6 to the fourth power, or 0.10368. The estimated horizon-four coefficient is -0.178410, with SE 0.101311. Its approximate 1.96 interval is [-0.3770, 0.0202]. This particular interval misses the true positive response. A coverage statement concerns repeated samples, so a particular estimated response can have the wrong sign and a particular interval can miss the truth. The displayed result has not been replaced with its theoretical target. The independent shock is known by construction in this experiment. Observed macroeconomic work still needs a credible shock measure, appropriate dated controls and justified inference.

Next, use repeated units to construct another comparison

Panel methods exploit changes within an identified unit.

> Time-series methods studied a single ordered history and its dynamics.

> The next lesson will derive first differences and fixed effects, then build a two-group policy comparison over time.